Experiments, Simulation and Descent

A Structural Derivation of Monotonicity, Fisher Geometry, Resolution Development and the Lorentz Bridge

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Abstract

We give one structural derivation, in the style of Čencov's characterization, of the facts that RG has so far assembled from separate arguments.

- Why operational information is monotone.
- What equality of information means.
- Why the space of experiments carries a canonical topology and order.
- Why the local geometry is Fisher.
- When geometry descends to observational quotients.
- When information develops monotonically.
- Why the Lorentz bridge has the form it has.

The central object. An experiment is represented as a distribution of posterior beliefs with barycentre equal to the prior. Garbling becomes conditional averaging. The simulation preorder then becomes convex order and, equivalently, comparison by all decision tasks (Blackwell–Sherman–Stein). The equality case is recoverability by one parameter-independent map.

The category path. Garblings form a category $\mathbf{Exp}_\Theta$. Lossless morphisms are exactly the congruent splittings. The posetal reflection of $\mathbf{Exp}_\Theta$ is isomorphic to the poset of Bayes-plausible measures under convex order. Mutual simulation equals equality of posterior laws, and each class has a canonical minimal representative.

The topology. The quotient of bounded experiments is compact Hausdorff, simulation is a closed order, and Le Cam's distance metrizes the quotient topology. The last fact is proved directly, by a compact-to-Hausdorff argument.

The local layer. Scores are conditional expectations, so Fisher is Loewner-monotone. Invariance under lossless morphisms forces $c\cdot$Fisher (Čencov). The non-idealized versions are quantitative stability $e^{\pm 2\varepsilon}$ under approximate invariance, and Fisher continuity under Le Cam closeness *with* derivative control (and its failure without).

Development. Information increases along observation filtrations. The increments are squared martingale differences, and the limit is taken in L^2. Physical time is monotone only for dynamics that fix a reference law.

The Lorentz bridge. For a cone experiment, scale invariance of the support forces log-homogeneity. The radial reflection of the positive Fisher form then has Lorentz inertia. The reflected form is quadratic (direction-independent) exactly for rank-two symmetric cones, the spin factors. The qubit state cone is one of them.

Evidence. All proofs are given or the theorems are imported with citations. Seven exact witnesses check constants; no claim rests on sampling.

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1 Scope and claim discipline

- **Classical and imported.** Blackwell–Sherman–Stein, Le Cam's randomization criterion, Čencov's characterization, Doob's martingale convergence, the gluing of basic tensors, and the Faraut–Korányi classification of symmetric cones and their characteristic functions.
- **Proved here.** Every step not marked as imported, including the RG-specific statements:
  - lossless morphisms are congruent splittings (from the Čencov print);
  - the localization description of the quotient;
  - metrization of the quotient by Le Cam's distance;
  - the non-idealized Čencov statements;
  - the cone dilation lemma and the rank-two characterization of the quadratic Lorentz bridge.
- **Novelty.** No priority is claimed for the assembly. The contribution is a single derivation in which every hypothesis is visible.

Assumption 1.1 (Standing assumptions)**.** (A1) *Finite parameters.* Θ is finite, with a strictly positive prior π. Sections 6–9 replace this by a smooth parameter manifold, stated there.

(A2) *Finite outcomes.* Experiments have finitely many outcomes. Section 5 bounds their number by N, padding with zero-probability outcomes.

(A3) *Garbling.* Garblings are parameter-independent Markov kernels.

(A4) *Dominated case.* Where continuous outcomes appear (cone experiments), they are dominated by Lebesgue measure.

2 The category of experiments

Definition 2.1 ($\mathbf{Exp}_\Theta$)**.** Objects are experiments $E = (X, \{P_\theta\}_{\theta\in\Theta})$ with finite X. A morphism $K : E \to F$ is a Markov kernel with $KP_\theta = Q_\theta$ for every θ; this is a *garbling*. Composition is the composition of kernels.

Definition 2.2 (Simulation preorder, lossless morphisms)**.**

- $E \succeq F$ iff a morphism $E \to F$ exists.
- $E \sim F$ iff $E \succeq F$ and $F \succeq E$.
- A morphism is *lossless* if it has a left inverse in the category of Markov kernels.

Lemma 2.3 (Lossless morphisms are congruent splittings; Čencov print, Lemma 3.2)**.** *A Markov kernel has a Markov left inverse iff its columns have pairwise disjoint supports. Equivalently, after removing unused outputs it is a congruent splitting composed with a relabeling.*

Definition 2.4 (Minimal representative)**.** For E, merge all outcomes with identical likelihood-ratio vectors, i.e. identical posteriors, and delete null outcomes. Call the result $E^{\min}$. The merge $E \to E^{\min}$ is a garbling. The kernel sending each merged outcome back to its constituents in proportion to their probability, which does not depend on θ because the likelihood ratios agree, is a lossless morphism $E^{\min} \to E$.

3 Posterior representation and the structural monotonicity theorem

Definition 3.1 (Posterior law)**.** For E and prior π, let $m_x = \sum_\theta \pi_\theta P_\theta(x)$ and $b_x(\theta) = \pi_\theta P_\theta(x)/m_x$ for $m_x > 0$. The *posterior law* is $\mu_E = \sum_x m_x \delta_{b_x}$ on the simplex $\Delta(\Theta)$.

Lemma 3.2 (Barycentre and reconstruction)**.** $\int b\, d\mu_E(b) = \pi$. *Conversely, every finitely supported μ on $\Delta(\Theta)$ with barycentre π (*Bayes plausible*) is μ_E for the experiment $P_\theta(b) = \mu(\{b\})\, b(\theta)/\pi_\theta$.*

Proof. $\sum_x m_x b_x(\theta) = \sum_x \pi_\theta P_\theta(x) = \pi_\theta$. For the converse, $\sum_b P_\theta(b) = \pi_\theta^{-1} \int b(\theta)\, d\mu = 1$, and Bayes' rule returns b with predictive mass $\mu(\{b\})$. □

Theorem 3.3 (Garbling is conditional averaging)**.** *If $F = KE$, then under the joint predictive law of (X, Y), $b_Y = \mathbb{E}[b_X \mid Y]$. Hence $\mu_F \preceq_{\mathrm{cx}} \mu_E$, i.e. $\int \Phi\, d\mu_F \le \int \Phi\, d\mu_E$ for every convex continuous Φ.*

Proof.

$$n_y c_y(\theta) = \pi_\theta Q_\theta(y) = \sum_x \pi_\theta P_\theta(x) K(y|x) = \sum_x m_x K(y|x)\, b_x(\theta),$$

and $m_x K(y|x)/n_y$ is the conditional law of X given $Y = y$. Then apply conditional Jensen. Witness: V1. □

Theorem 3.4 (Representation; Blackwell–Sherman–Stein [1, 2, 3])**.** *For finite experiments and a full-support prior, the following are equivalent:*

(i) $E \succeq F$*;*

(ii) $\mu_F \preceq_{\mathrm{cx}} \mu_E$*;*

(iii) $V_U(E) \geq V_U(F)$ *for every finite decision problem* U*, where*

$$V_U(E) = \int \max_a \sum_\theta b(\theta)U(\theta, a)\, d\mu_E(b).$$

Proof. *(i)⇒(ii).* This is Theorem 3.3.

(ii)⇒(iii). $\Phi_U(b) = \max_a \sum_\theta b(\theta)U(\theta, a)$ is convex.

(iii)⇒(ii). Every convex piecewise-linear Φ on the simplex is some Φ_U, and these are dense in the convex continuous functions on a compact convex set.

(ii)⇒(i). Write $\mu_E = \sum_i m_i \delta_{b_i}$ and $\mu_F = \sum_j n_j \delta_{c_j}$. We need $\gamma_{ij} \geq 0$ with $\sum_j \gamma_{ij} = m_i$ and $\sum_i \gamma_{ij} b_i = n_j c_j$. If this system is infeasible, Farkas' lemma gives scalars a_i and vectors d_j with $a_i + d_j \cdot b_i \geq 0$ for all i, j and $\sum_i m_i a_i + \sum_j n_j d_j \cdot c_j < 0$. Put $\Phi(b) = \max_j(-d_j \cdot b)$, which is convex. Then

$$\int \Phi\, d\mu_E \leq \sum_i m_i a_i < -\sum_j n_j d_j \cdot c_j \leq \int \Phi\, d\mu_F,$$

contradicting (ii). Hence γ exists. Set $K(j|i) = \gamma_{ij}/m_i$. Lemma 3.2 gives $\sum_i K(j|i)P_\theta(i) = \sum_i \gamma_{ij} b_i(\theta)/\pi_\theta = n_j c_j(\theta)/\pi_\theta = Q_\theta(j)$. Witness: V2, where the LP returns K with $KP = Q$ exactly. □

Corollary 3.5 (Why monotonicity is everywhere)**.** *A functional on experiments is monotone under all garblings iff it is monotone under convex order of posterior laws. In particular:*

- *every* $\int \Phi\, d\mu_E$ *with* Φ *convex is monotone;*
- *every decision value* V_U *is monotone;*
- *mutual information* $I_\pi(\Theta; X) = \int D(b\|\pi)\, d\mu_E$ *is monotone;*
- *every* f*-divergence between two laws is monotone, since it equals* $\int \Phi\, d\mu$ *for a two-point prior and a convex* Φ*.*

Monotonicity is not a property of each formula. It is the image of one order.

4 Equality is recovery

Theorem 4.1 (Equality case)**.** *Let* $F = KE$ *and let* Φ *be strictly convex. The following are equivalent:*

(i) $\int \Phi\, d\mu_F = \int \Phi\, d\mu_E$*;*

(ii) $b_X = b_Y$ *almost surely;*

(iii) the reverse kernel $R(x|y) = m_x K(y|x)/n_y$ *satisfies* $RKP_\theta = P_\theta$ *for every* θ*;*

(iv) $E \sim F$*.*

Proof. *(i)⇔(ii).* Equality in conditional Jensen for strictly convex Φ holds iff b_X is Y-measurable, so $b_X = \mathbb{E}[b_X \mid Y] = b_Y$.

(ii)⇒(iii).

$$(RKP_\theta)(x) = \sum_y \frac{m_x K(y|x)}{n_y} Q_\theta(y) = m_x \sum_y K(y|x) \frac{c_y(\theta)}{\pi_\theta} = \frac{m_x b_x(\theta)}{\pi_\theta} = P_\theta(x).$$

(iii)⇒(iv)⇒(i). R is a morphism $F \to E$, and convex order is antisymmetric (Proposition 4.3). Witness: V3, where a merge of outcomes with equal posteriors loses no mutual information and the reverse kernel recovers every law exactly, while a lossy merge loses 0.024 nats. □

Corollary 4.2 (Mutual information detects strict loss, not order)**.** *Along a given garbling, $I_\pi(\Theta; X) - I_\pi(\Theta; Y) \geq 0$, with equality iff the garbling is recoverable. Any single scalar can still fail to order incomparable experiments. The order is carried by all decision values jointly.*

Proposition 4.3 (Antisymmetry; the canonical form)**.** *$\mu \preceq_{\text{cx}} \nu$ and $\nu \preceq_{\text{cx}} \mu$ imply $\mu = \nu$. Hence $E \sim F$ iff $\mu_E = \mu_F$ iff $E^{\min} \cong F^{\min}$ up to relabeling.*

Proof. Equality of $\int \Phi$ for all convex Φ extends to differences of convex functions, which include all polynomials and are therefore dense in $C(\Delta(\Theta))$. A measure is determined by its integrals against a dense set. $E^{\min}$ has one outcome per atom of μ_E, so μ determines $E^{\min}$ up to relabeling. □

Theorem 4.4 (Posetal reflection and localization)**.** *(a) The posetal reflection $\mathbf{Exp}_\Theta/\!\sim$ is isomorphic, via $E \mapsto \mu_E$, to the poset of finitely supported Bayes-plausible measures under convex order.*

(b) Two experiments are equivalent iff they are connected by a zigzag of lossless morphisms through their common minimal representative.

(c) Any functor or functional invariant under lossless morphisms factors through $\mathbf{Exp}_\Theta/\!\sim$. This is RG's descent.

Proof. (a) follows from Theorem 3.4, Lemma 3.2 (surjectivity) and Proposition 4.3 (injectivity). (b) follows from the minimal-representative construction and Proposition 4.3. (c) holds because invariance under the zigzag of (b) means constancy on $\sim$-classes. □

5 Topology of informational content

Definition 5.1. $\mathcal{E}_N = \prod_{\theta \in \Theta} \Delta_N$, with experiments padded to N outcomes. It is compact.

Definition 5.2 (Le Cam deficiency and distance)**.** The deficiency is

$$\delta(E, F) = \min_K \max_\theta \mathrm{TV}(KP_\theta, Q_\theta),$$

where K ranges over the compact set of $N \times N$ Markov kernels. The distance is $\Delta(E, F) = \max\{\delta(E, F), \delta(F, E)\}$.

Theorem 5.3 (Compact Hausdorff quotient, closed order, metrization)**.** *(a) δ is continuous on $\mathcal{E}_N \times \mathcal{E}_N$. Moreover $\delta(E, F) = 0$ iff $E \succeq F$.*

(b) Δ is a metric on $\mathcal{E}_N/\!\sim$.

(c) The quotient topology on $\mathcal{E}_N/\!\sim$ coincides with the Δ-topology. In particular the quotient is compact Hausdorff.

(d) Simulation is a closed relation: if $E_k \to E$, $F_k \to F$ and $E_k \succeq F_k$, then $E \succeq F$.

Proof. (a) $(E, F, K) \mapsto \max_\theta \mathrm{TV}(KP_\theta, Q_\theta)$ is continuous, and minimizing over a compact set preserves continuity. The minimum is attained, so $\delta = 0$ gives an exact K.

(b) Symmetry and the triangle inequality follow from composing near-optimal kernels: deficiencies are subadditive under composition, because TV contracts under kernels. $\Delta = 0$ iff $E \sim F$, by (a).

(c) By (a), Δ is continuous on $\mathcal{E}_N \times \mathcal{E}_N$ and constant on classes. So the identity map from the quotient topology to the Δ-topology is continuous. A continuous bijection from a compact space to a Hausdorff (metric) space is a homeomorphism.

(d) Take kernels K_k with $K_k E_k = F_k$. Pass to a convergent subsequence $K_k \to K$, by compactness. By continuity, $KE = F$. □

Imported Theorem 5.4 (Le Cam's randomization criterion [4, 5])**.** *$\delta(E, F) \le \varepsilon$ iff $V_U(F) \le V_U(E) + \varepsilon \|U\|_\infty$ for every bounded decision problem with finitely many actions. This is the* non-idealized *Blackwell theorem: approximate simulation is approximate dominance in every task.*

Remark 5.5 (What the topology does and does not do). The order comes from simulation. The topology makes its limits, continuity and quotient precise. A topology alone distinguishes no "more" from "less".

6 Local geometry: monotone Fisher, Čencov, and the non-idealized case

Assumption 6.1 (Smooth regular families)**.** $\theta \in M$, an open subset of $\mathbb{R}^d$. Each P_θ is positive on a finite X and C^1 in θ. The score is $S = \nabla \log P_\theta$, and Fisher is $F = \mathbb{E}[SS^\top]$.

Lemma 6.2 (Score descent)**.** *If $Q_\theta = KP_\theta$, then $S_Y = \mathbb{E}[S_X \mid Y]$ and*

$$F_X - F_Y = \mathbb{E}\big[\,\mathrm{Cov}(S_X \mid Y)\big] \succeq 0.$$

Proof. $\nabla Q_\theta(y) = \sum_x K(y|x) P_\theta(x) S_X(x)$; divide by $Q_\theta(y)$. Then use the law of total covariance. □

Corollary 6.3 (Global order implies local order)**.** *$E \succeq F$ implies $F_E(\theta) \succeq F_F(\theta)$ (Loewner) for every θ.*

No-Go 6.4 (Local order does not imply global order). Bernoulli($\sin^2 \frac{\theta}{2}$) and a pair of i.i.d. Bernoulli($\sin^2 \frac{\theta}{2\sqrt{2}}$) have equal Fisher information ($\equiv 1$) at every θ. Their Hellinger affinities at separation 1 differ (0.8776 against 0.8801), so neither simulates the other. See the Čencov print, No-Go 9.3.

Imported Theorem 6.5 (Čencov from descent; Čencov print, Theorem 3.4 [10])**.** *A family of positive quadratic forms on all finite simplices that is continuous and invariant under lossless morphisms (Lemma 2.3) equals c·Fisher, with one constant in every dimension.*

Structural reading. By Theorem 4.4(c), invariance under lossless morphisms is exactly descent to **Exp**/$\sim$. Čencov's hypothesis is therefore not an idealization imposed from outside. It is the statement that the metric is a function of informational content, which every RG quantity must be.

Theorem 6.6 (Non-idealized Čencov)**.** *(a)* Approximate descent. *If the invariance holds up to factors $e^{\pm\varepsilon}$ under every lossless morphism, then $h = cF$ up to $e^{\pm 2\varepsilon}$ everywhere. Without a separate permutation hypothesis the bound is $e^{\pm 4\varepsilon}$ (Čencov print, Theorems 4.3–4.4).*

(b) Le Cam closeness with derivative control. *If $\Delta(E_k, E) \to 0$ and the square-root Jacobians converge with a bound $\|J\| \le M$ and $\|\tilde{J} - J\| \le \eta$, then $\|F_k - F\| \le 4\eta(2M + \eta)$ (Čencov print, Theorem 10.4).*

(c) Without derivative control, *Le Cam closeness does not control Fisher: $\Delta \le 1/n$ while $F_n(0) = 4n^2$ (Čencov print, No-Go 10.3).*

Why this "un-idealizes" Čencov. The exact theorem needs exact invariance. Theorem 6.6 shows that approximate invariance gives approximately Fisher, with explicit constants, and that the operational topology (Le Cam) controls Fisher precisely when derivatives are declared. No sampling enters at any step.

Imported Theorem 6.7 (Hessian route; Hessian print, Theorem 3.1)**.** *A divergence that is nonnegative, C^2, nondegenerate and monotone under all garblings has Hessian $c \cdot F$. By Corollary 3.5, every f-divergence is such a divergence.*

7 Descent of geometry to observational quotients

Theorem 7.1 (Basic tensors descend)**.** *Let $\varpi : M \to B$ be a surjective submersion with connected fibres and vertical distribution $V = \ker D\varpi$. A symmetric tensor g on M is the pullback of a tensor on B iff it is* basic*, i.e. for every vertical field W,*

$$\iota_W g = 0 \qquad \text{and} \qquad \mathcal{L}_W g = 0.$$

Proof. Necessity holds because pullbacks are annihilated by vertical vectors and invariant along fibres. For sufficiency, on horizontal lifts g depends only on the projected vectors by $\iota_W g = 0$. By $\mathcal{L}_W g = 0$ it is constant along connected fibres, so it defines a tensor on B. This is standard [6]. □

Corollary 7.2 (Statistical tensors descend; protocol tensors must be checked)**.**

- *On a regular constant-rank region where the law map p factors as $M \to p(M)$ with $p(M)$ an embedded submanifold, the pullback p^*F is basic and equals the pullback of the restricted Fisher metric of $p(M)$.*
- *A separately proposed tensor must satisfy both conditions. For example, $(1 + y^2)\,dx^2$ annihilates ∂_y but is not invariant along it, so it does not descend through $(x, y) \mapsto x$ (V4).*

No-Go 7.3 (Singular points)*.* At singular points $\ker Dp$ need not be an exact fibre. The family $p_\theta = (\frac{1}{2} + \theta^2, \frac{1}{2} - \theta^2)$ has $F(0) = 0$ but distinct laws for every $\theta \neq 0$. Descent there is typed by the walls of the print (L129–L131) and by rate invariants (Rooting Closure).

8 Resolution development

Theorem 8.1 (Filtration development)**.** *Let a fixed regular experiment have square-integrable score S, and let $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \cdots$ be observation σ-fields generated by parameter-independent functions of the record. Put $S_n = \mathbb{E}[S \mid \mathcal{G}_n]$ and $F_n = \mathbb{E}[S_n S_n^\top]$. Then*

$$F_{n+1} - F_n = \mathbb{E}\big[(S_{n+1} - S_n)(S_{n+1} - S_n)^\top\big] \succeq 0,$$

and $S_n \to \mathbb{E}[S \mid \mathcal{G}_\infty]$ in L^2, so $F_n \to F_\infty$.

Proof. $\mathbb{E}[S_{n+1} \mid \mathcal{G}_n] = S_n$, so the increment is orthogonal to S_n in L^2. Expanding gives the identity. The convergence is the L^2 martingale convergence theorem [7]. □

Corollary 8.2 (Posterior development)**.** *The posterior laws μ_n of $\mathcal{G}_n$ increase in convex order. The posteriors form a bounded martingale, which converges almost surely and in law (Doob), so $\mu_n \to \mu_\infty$ weakly.*

Reading. Refinement increases identifiable information by an exact squared increment. A plateau means no new *local* score information; it need not mean no new global distinction. Independent repetition is different: there $F_n = nF_1$ diverges.

Theorem 8.3 (When physical time is monotone)**.** *(a) If T_t is a Markov semigroup with $T_t\pi = \pi$, then $D(T_{t+s}p\|\pi) \le D(T_tp\|\pi)$.*

(b) For a reversible finite chain with rates q_{ij} and $p_i = \pi_i f_i$,

$$\frac{d}{dt}D(p_t\|\pi) = -\tfrac{1}{2}\sum_{i,j}\pi_i q_{ij}(f_j - f_i)(\log f_j - \log f_i) \le 0,$$

with equality, under irreducibility, only at π (V7).

(c) An invertible evolution of the complete record has no strict information loss. A periodic trajectory carries no strictly monotone state function.

Proof. (a) T_s is a garbling fixing π; apply Corollary 3.5. (b) is the symbolic identity V7; each summand is nonnegative. (c) Processing in both directions forces equality, and periodicity forces equal values. □

9 The Lorentz bridge, structurally

Definition 9.1 (Cone experiment)**.** Let $C \subset \mathbb{R}^d$ be an open convex cone, regular (containing no line), with dual cone C^*. The *cone experiment* has laws $p_z(s) = Z(z)^{-1}e^{-\langle z,s\rangle}\mathbf{1}_C(s)$ for z in the interior of C^*, where $Z(z) = \int_C e^{-\langle z,s\rangle}ds$ is the Koszul–Vinberg characteristic function.

Lemma 9.2 (Dilation forces log-homogeneity)**.** *$Z(\lambda z) = \lambda^{-d}Z(z)$. Hence $\Psi = \log Z$ is log-homogeneous: $\Psi = -\frac{d}{2}\log q$ with $q = Z^{-2/d}$, which is two-homogeneous and positive. Its Fisher form $F = \operatorname{Hess}\Psi$ is positive definite.*

Proof. Substitute $s = s'/\lambda$. The cone is invariant, $C/\lambda = C$, and $ds = \lambda^{-d}ds'$. The Fisher form is the covariance of the sufficient statistic s, which is positive because C has nonempty interior. Witness: V5, exact for $d = 2$ and $d = 3$. □

Theorem 9.3 (Radial reflection; companion protocol, Theorem 4.2)**.** *With $\alpha = d/2$,*

$$g = F - \alpha^{-1}d\Psi \otimes d\Psi = -\frac{\alpha}{q}\operatorname{Hess} q$$

has inertia $(1, d-1)$. The radial direction is the negative one, and its F-orthogonal complement is positive.

Proof. Euler's identity gives $d\Psi(z) = -2\alpha$ and $F(z,\cdot) = -d\Psi$. Hence $F(u,u) = 1$ for $u = z/\sqrt{2\alpha}$, and $g = F - 2F(u,\cdot)^{\otimes 2}$. In the decomposition $v = au + w$ with $F(u,w) = 0$, we get $g(v,v) = -a^2 + F(w,w)$. Direct differentiation of $-\alpha\log q$ gives the second expression. □

Where monotone structure enters. The $d-1$ positive directions of g are exactly Fisher restricted to the complement of the dilation generator. Their positivity is forced by Lemma 6.2, Čencov, and the covariance argument. The single negative direction is the dilation generator, which is supplied by the cone's scale invariance (Lemma 9.2). It is *not* supplied by monotonicity.

Imported Theorem 9.4 (Symmetric cones [8, 9])**.** *Every symmetric cone is a product of irreducible ones, namely the Lorentz cones and the cones of positive matrices over* $\mathbb{R}$*,* $\mathbb{C}$*,* $\mathbb{H}$ *(and one exceptional cone). For an irreducible symmetric cone of rank* r *and dimension* d*,* $Z(z) = c \det(z)^{-d/r}$*, where* det *is the Jordan determinant, a homogeneous polynomial of degree* r*.*

Theorem 9.5 (The quadratic Lorentz bridge is the rank-two case)**.** *For an irreducible symmetric cone,* $q = Z^{-2/d} \propto \det^{2/r}$*. Hence:*

(a) q *is a quadratic form, so* $g = -(\alpha/q) \operatorname{Hess} q$ *has a direction-independent Lorentz form, iff* $r = 2$*, i.e. iff* C *is a Lorentz cone (spin factor);*

(b) for $r \geq 3$*,* g *is still of Lorentz inertia by Theorem 9.3, but* $\operatorname{Hess} q$ *varies with direction, a Lorentz–Finsler form. For example, for* 3×3 *positive matrices* $q = \det^{2/3}$ *has a non-constant Hessian (V6);*

(c) the qubit state cone, 2×2 *Hermitian positive matrices, has rank* 2 *and dimension* 4*. It is the* $1 + 3$ *Lorentz cone, with* $\det(tI + x \cdot \sigma) = t^2 - |x|^2$*.*

Proof. (a) $\det^{2/r}$ is a polynomial of degree 2 iff $r = 2$. For $r = 2$, the Jordan determinant of a spin factor is the Lorentz quadratic form. (b) $\det^{2/3}$ is not a polynomial, and V6 checks it directly. (c) Expand the determinant. □

Corollary 9.6 (Lorentz-group invariance forces quadraticity)**.** *If the experiment's symmetry group contains the identity component of a Lorentz group acting on* $\mathbb{R}^{1+n}$ *(*$n \geq 2$*), with the cone experiment equivariant (*$p_{\Lambda z} = \Lambda_*^{-\top} p_z$*, a lossless re-description by a volume-preserving bijection), then* q *is Lorentz-invariant and two-homogeneous, hence proportional to* $t^2 - |x|^2$*. The Fisher form is invariant under the same group acting on parameters, which is consistent with the compact stabilizer of the companion's No-Go 5.4.*

The bridge in one line:

- descent forces positive Fisher;
- the cone's scale invariance supplies one distinguished direction;
- radial reflection yields Lorentz inertia;
- rank two, or Lorentz invariance, makes the form quadratic.

The physical identification of the dilation generator with a clock remains a declaration.

10 The organizing theorem and its assumption ledger

Organizing Theorem. *Under Assumption 1.1 and the smoothness assumptions of Section 6:*

(I) Simulation $\iff$ *convex order of posterior laws* $\iff$ *dominance in every decision task (Theorem 3.4). Every operational information functional is monotone (Corollary 3.5).*

(II) Equality under a strictly convex functional holds iff the garbling is recoverable by one parameter-independent map (Theorem 4.1).

(III) The posetal reflection is the poset of Bayes-plausible measures. Descent is factorization through it (Theorem 4.4). The quotient is compact Hausdorff, metrized by Le Cam's distance, with a closed order (Theorem 5.3).

(IV) Locally, garbling contracts Fisher, and descent selects c·Fisher. Approximate descent gives $e^{\pm 2\varepsilon}$, and Le Cam closeness with derivative control gives Fisher closeness (Theorem 6.6).

(V) Basic tensors, and only they, descend to regular observational quotients (Theorem 7.1).

(VI) Observation filtrations develop information monotonically, with an L^2 limit. Physical time is monotone only for dynamics that fix a reference law (Theorems 8.1, 8.3).

(VII) Cone experiments carry a Lorentz-signed reflection of their positive Fisher form. It is quadratic exactly for rank-two symmetric cones, which include the qubit cone (Theorems 9.3, 9.5).

	Needs	Does not need
I	finite Θ, finite outcomes, full-support prior	any metric, smoothness, Fisher
II	a strictly convex Φ	a choice among such Φ
III	bounded outcome number N	sampling; any metric beyond TV
IV	smoothness, positivity, continuity (exact); a derivative bound (non-ideal)	monotonicity under *all* channels: descent suffices
V	regular region, connected fibres	global regularity
VI	parameter-independent refinements; or $T_t\pi = \pi$ for time	that physical time is a data-processing order
VII	a cone-shaped support (scale invariance); rank two for quadraticity	a derivation of spacetime; the clock identification is declared

11 The category path in one diagram

$$\mathbf{Exp}_\Theta \xrightarrow{\text{posetal reflection}} \mathbf{Exp}_\Theta/\!\sim \xrightarrow[\cong]{\mu} (\mathcal{B}_\pi, \preceq_{\mathrm{cx}}) \xrightarrow{(V_U)_U} \prod_U \mathbb{R}.$$

- The first arrow inverts the lossless morphisms (congruent splittings).
- The second is the posterior representation, an isomorphism of posets.
- The third is the faithful decision profile, which is a topological embedding on $\mathcal{E}_N/\!\sim$.

Local geometry is the functor $E \mapsto F_E(\theta)$, which is monotone along garblings and constant on isomorphism classes. Čencov's theorem says that, up to scale, it is the only such functor of quadratic forms.

12 Exact witnesses and limits

- `rg_structural_witnesses2.py` runs V1–V7, exact or symbolic, with no sampling. Reruns are byte-identical.
  - V1: conditional averaging.
  - V2: LP reconstruction of a garbling from convex order.
  - V3: the equality case and exact recovery.
  - V4: a non-basic tensor.

  - V5: cone dilation in two and three dimensions.
  - V6: the non-quadratic rank-three q.
  - V7: the reversible dissipation identity.

- *Limits.*
  - The classical case only; the quantum analogue of Theorem 3.4 (quantum Blackwell theorems) is not developed here.
  - The infinite-outcome case is restricted to dominated cone experiments.
  - The imported theorems are cited.
  - The continuity question of the Čencov print is inherited.

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